DEPARTMENT OF MATHEMATICS
UNIVERSITY COLLEGE OF ENGINEERING (A) – O.U., HYD - 7
B.E. (CSE) III- SEMESTER, CLASS TEST- II
Engineering Mathematics-III
(NUMERICAL METHODS)

TIME: **1:30 hours** **Max. Marks: 30**
(Academic Year: 2025-2026) **Date: 24-10-2025**

			Marks	CO	BT
	PART – A **10** (Answer **all questions**)	**5×2 =**			
1)a.	The following data represents e^{-x}		**2**	**4**	**L2**

x	-1	-0.5	0	1
f(x)	2.7183	1.6487	1	0.3679

Obtain an approximate value of $f''(-1)$ using the method $f''(x_0) = [f_0 - 2f_1 + f_2]/h^2$, with $h = 1$.

b.	Find the second divided difference of $f(x) = \frac{1}{x}$, using the points x_0, x_1, x_2.	**2**	**4**	**L2**
c.	Evaluate $\int_0^1 \frac{dx}{3+2x}$ using Trapezoidal rule with $n = 10$.	**2**	**4**	**L3**
d.	Evaluate $y(0.1)$ by Taylor's series method if $y(x)$ satisfies $y' = xy + 1$, $y(0) = 1$.	**2**	**5**	**L2**
e.	Apply Euler's method to solve $y' = x + y$, $y(0) = 0$ choosing the step length $h = 0.2$ (Carry out 6 steps).	**2**	**5**	**L3**
	PART - B (Answer any **TWO** questions) **2×10= 20**			
2	a. Find the number of men getting wages between ₹100 and ₹ 150 from the following data:	**5**	**4**	**L3**

Wages in ₹	0-100	100-200	200-300	300-400
Frequency	9	30	35	42

b. Given that

x	1.0	1.1	1.2	1.3	1.4	1.5	1.6
y	7.989	8.403	8.781	9.129	9.451	9.750	10.031

Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at $x=1.1$ and $x=1.6$. | **5** | **4** | **L3** |

3	a. Using Picard's method, solve $\frac{dy}{dx} = -xy$, $y(0) = 1$ up to third approximation.	**5**	**5**	**L3**
	b. Apply Runge-Kutta method to find an approximate value of y for $x=0.2$ *in steps of 0.1*, if $\frac{dy}{dx} = x + y^2$, $y(0) = 1$.	**5**	**5**	**L4**
4	a. Given the values	**5**	**4**	**L3**

X:	5	7	11	13	17
f(x):	150	392	1492	2366	5202

Evaluate $f(9)$ using Lagrange's formula.
b. Use Runge-Kutta method of fourth order , solve for y at $x=1.2$ from $\frac{dy}{dx} = \frac{2xy + e^x}{x^2 + xe^x}$, $y(0) = 1$. | **5** | **5** | **L3** |